

RESEARCH ARTICLE

Challenges and Perspectives in Interpretable Music Auto-Tagging Using Perceptual Features

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This work was supported by the Framework of H.F.R.I call “Basic Research Financing (Horizontal support of all Sciences)” through the National Recovery and Resilience Plan “Greece 2.0” funded by the European Union–NextGenerationEU (H.F.R.I. Project-Emotional Artificial Intelligence in Music Expression) under Grant 15111. The publication of the article in OA mode was financially supported by HEAL-Link.

ABSTRACT In the era of music streaming platforms and recommendation systems, the automatic music auto-tagging task has gained a lot of traction and has motivated researchers to develop methods for solving the task focusing on improving performance metrics on baseline datasets. The majority of recent approaches rely on deep neural networks, which despite their impressive performance, are opaque, meaning it is difficult to explain their output on a given input. While the problem of interpretability has been highlighted in other domains, such as medicine, it has not been a priority for music-related tasks. In this work, we explored the usefulness of interpretability for music auto-tagging. We developed a pipeline incorporating three types of information extraction procedures: 1) symbolic knowledge, 2) auxiliary deep neural networks, and 3) signal processing, to extract perceptual features of audio files, which were then used to train an explainable machine learning model to predict tags. We experimented on three datasets the *MTG-Jamendo* dataset, the *GTZAN* dataset, and the *MagnaTagATune* dataset. Our method outperforms baseline models in all tasks and in some cases is competitive with the state-of-the-art. We conducted a human survey to evaluate user trust in our methodology and in a state-of-the-art model, concluding that while the state-of-the-art model offers better performance, there are use cases where the slight deterioration in accuracy is outweighed by the increased trust and value provided by interpretability.

INDEX TERMS Music understanding, explainable AI, perceptual musical features.

I. INTRODUCTION

Music auto-tagging is an umbrella term for tasks that involve assigning music-related tags to songs and can be especially useful for organizing large collections of music. These tags are related to musically meaningful information such as instruments [22], genres [15], mood [63], emotions [47], harmony [12], rhythm [61], and other metadata [43]. Most approaches for solving such tasks rely on feature extraction and machine learning (ML), where recently the feature extraction step is minimal, such as focusing solely on

extracting Mel-frequency cepstral coefficients (MFCCs), with a growing trend toward deep learning models [18] and foundation models [40]. This tendency is accompanied by models becoming more opaque and difficult to explain, which can be especially problematic in cases where the ground truth itself can be subjective and ambiguous [8], such as emotion and mood tags.

In these cases, the reliability of the training data dictates the reliability of the model, and without interpretability, it is difficult to detect flaws or biases in either data or model [55]. For music especially, where real-world data tend to be diverse, contrary to available datasets [34], it is important to be able to detect and fix biases in order

The associate editor coordinating the review of this manuscript and approving it for publication was Manuel Rosa-Zurera.

to make robust systems that can be applied in real-world settings. As an extreme example, consider a mood tagging system trained exclusively on pop and rock music. Such a system is unlikely to generalize well to other genres, such as classical music, and its behavior on unfamiliar inputs would be unpredictable. Additionally, end-users may not be aware of the training data distribution or the specifics of machine learning models, which limits the system's practical applicability, even if it performs well on test data. However, if the system were interpretable, users could better understand its tagging decisions, potentially increasing trust and usability in real-world settings, and assuming satisfactory performance.

Note that the notions of interpretability and explainability, are often used interchangeably, and there are many, sometimes conflicting definitions of the terms in the literatures [1], [38], and [42]. For instance, some consider attention mechanisms to be interpretable [69], while others consider them to be misleading [35]. In our work, we consider the definition of interpretability proposed in [17]: “interpretability as the ability to explain or present [the behavior of a machine learning model] in understandable terms to a human.”, which refers to analyzing both why and how a model predicts. Machine learning models may inherently possess such an ability (e.g. XGBoost) or may require auxiliary methods (e.g. SHAP) to acquire it [49]. However, we loosely use the terms *explanations* and *interpretations* interchangeably.

In recent years, there have been some efforts to mitigate the interpretability issue in music auto-tagging systems by extracting perceptual musical features that are understandable to humans [2], [13], [14], [24], [46]. Perceptual features are characteristics that have musical and acoustic meaning, such as tonality, rhythmic stability, and loudness, and are highly correlated with emotions [25], [66]. Because of this correlation, these features perform well in Music Emotion Recognition (MER) tasks. These characteristics are derived from music concepts and theoretical knowledge and as described in [30] there are few quantitative works on them. In our work, we build on these ideas and implement a pipeline for interpretable music auto-tagging (see Figure 1), utilizing perceptual features.

Specifically, our contributions can be summarised as follows:

- Three distinct approaches are integrated for extracting perceptual musical features: signal processing, deep neural networks, and symbolic knowledge representation.
- A novel set of harmony-related musical features is introduced by establishing a comprehensive music knowledge base and applying reasoning methodologies.
- Model interpretability is explored through five techniques: SHAP [45], XGBoost feature importance [11], permutation-based feature importance, ablation studies, and natural language explanations generated via prompting ChatGPT with SHAP values.
- Challenges in ML-based auto-tagging, such as uncovering dataset biases hidden in deep learning models and

enhancing the understanding of ambiguous labels, are addressed using these interpretability methods.

- A human evaluation is conducted to assess the trustworthiness and effectiveness of the interpretable approach in comparison to state-of-the-art models for music auto-tagging.

The paper is organized as follows: Section II reviews related work in the fields of Music Information Retrieval (MIR) and Explainable AI (XAI). Section III details the methodology used to extract perceptual musical features and perform interpretable music auto-tagging. Section IV presents the datasets used and the results of our approach. Section V covers our human evaluation survey, designed to assess both our interpretable model and a state-of-the-art model (SOTA). Finally, Section VI reflects on key insights from this work and suggests directions for future research.

II. RELATED WORK

Music auto-tagging involves categorizing music tracks based on various metadata, typically framed as a multilabel classification problem [20]. To address this challenge, several architectures have been developed, primarily focusing on the audio modality of the music. Convolutional networks, such as the VGG-ish model [32] and Musicnn [54], are widely used for extracting information from time-frequency representations of audio, while models like Wave-U-Net [60] and TCNN [51] are designed to process raw audio signals directly. Recently, the Audio Spectrogram Transformer (AST) has emerged as a promising solution for audio classification tasks [27], and a cutting-edge foundation model has been developed, capable of performing various tasks, including music auto-tagging [70]. Other existing architectures leverage different modalities; for example, MIDI2vec [43] focuses on graphical symbolic music, and the JMLA model [19] facilitates zero-shot music auto-tagging by integrating audio and language modalities. In our work, we focus on the audio modality of music, incorporating deep neural network architectures like VGG-ish, alongside signal processing techniques and symbolic AI systems, such as knowledge graphs, within our XAI framework.

Explainable AI has emerged as a vital research area aimed at enhancing the interpretability and transparency of machine learning models. By offering clear insights into the decision-making processes of these models, XAI techniques are essential for building user trust and enabling the responsible deployment of AI systems [52]. In [28], a comprehensive taxonomy of explainability methods is presented, categorizing them into two main types: opaque explanations and transparent-by-design models. Opaque explanations leverage post-hoc techniques to clarify the decisions made by opaque models, whereas transparent-by-design models are inherently interpretable due to their structural parameters. Our work integrates both approaches, employing SHAP values for post-hoc explainability and

utilizing the weights of XGBoost as part of the interpretable-by-design methodology.

In the music domain, XAI methods have been applied to enhance model interpretability through various approaches. Attention mechanisms, as demonstrated by [69], help highlight which parts of the musical input are most significant to a model's predictions, offering insights into the prioritization of specific musical elements. Furthermore, genre- and spectral-based prototypes [3], [44] seek to identify distinctive patterns across different music genres, connecting model decisions to concrete auditory features. Concept-based explanations [23] go further by linking latent model representations to higher-level human-understandable concepts, facilitating a more intuitive understanding of how models interpret and process musical data. In [59], the authors proposed MusicLIME, a model-agnostic method for explaining feature importance in music auto-tagging. Our approach focuses on perceptual musical features [46], aiming to extract human-interpretable characteristics and integrate them into inherently interpretable models. This manuscript builds upon and extends that prior work.

Several studies have examined the comprehensibility and effectiveness of explanations across a variety of intelligent systems. For example, one study explores the demand for and comprehensibility of explanations in low-cost Intelligent Interactive Systems [9], while another presents findings from a controlled study with over 200 participants, investigating the effectiveness of different types of explanations in intelligent systems [41]. Additionally, research has investigated user experiences with various explanation types in recommender systems [26]. Further, another study delves into how the complexity of decision sets impacts model interpretability, measured through human-simulatability tasks [39]. Moreover, recent work documents the prototyping, testing, and redesign cycles of an explainable AI technique and its user interface for clinical Decision Support Systems (DSS) [52]. In our research, we aim to measure the degree to which interpretable approaches enhance user trust and to identify the types of explanations that are most effective in achieving this.

III. METHODOLOGY

Our approach combines the extraction of perceptual musical features with the application of various XAI techniques to achieve interpretable music auto-tagging.

A. EXTRACTING PERCEPTUAL FEATURES

To extract perceptual features from the datasets, we employed three different approaches: Symbolic Knowledge, Deep Neural Networks, and Signal Processing.

1) SYMBOLIC KNOWLEDGE

By utilizing symbolic knowledge, we created a new set of interpretable features that derive from music harmony. To accomplish this, we constructed a knowledge base in

order to extract music features from each track's chords. Our knowledge base consists of chords as instances, axioms, classes and subclasses from the Functional Harmony Ontology (FHO) [36]. These classes and subclasses describe the harmonic function of each chord in a music track [67]. Using SPARQL queries, we retrieved the instances of certain classes to create useful features, such as the frequency of specific harmonic patterns (e.g., *dominant-dominant-tonic*), in the form of n-grams.

For implementation, we used audio to recognize the chords in each music track with Omnizart's Python library.¹ Our knowledge base was set up on GraphDB [29], and we utilized its SPARQL endpoint. In total, we created a 32-dimensional feature vector $x_{\text{harmonic}} \in \mathbb{R}^{32}$. First, we created *Dominants* and *Subdominants* by calculating the ratio between the sum of dominant and subdominant retrieved chords and the total number of chords, respectively. Additionally, by defining the function of a chord in any chord progression, we developed 30 more features based on bigrams and trigrams (e.g., *sub-dom*, *sub-sub-dom*). To construct these features, we also designated some chords as tonics - chords that follow a dominant without serving another function, and globs - major dominant chords that produce a great deal of harmonic tension.

2) DEEP NEURAL NETWORKS

We leveraged the power of deep neural networks (DNNs) to efficiently capture perceptual features from a dataset curated by music experts [2]. We first extracted MFCCs from the initial 30 seconds of each audio file. MFCCs represent a power spectrum that's scaled to approximate the human auditory system's frequency response. They are a matrix $K \subset \mathbb{R}^{S \times P}$, where S is the number of segments and P is the number of MFCCs.

To build our DNN model, we used a VGG-ish architecture trained with multiple linear regression. Given an input MFCC x , where $x \in K$, our DNN outputs a feature vector $x_{\text{midlevel}} \in \mathbb{R}^p$, where $p \in \mathbb{N}$ corresponds to the number of extracted features. Specifically, we trained our model on the *Mid Level Perceptual Music Features* dataset, which contains 5000 songs in MFCC format. Each song is annotated by music experts with seven perceptual features: *Melodiousness*, *Articulation*, *Rhythmic Stability*, *Rhythmic Complexity*, *Dissonance*, *Tonal Stability*, and *Minorness* [2]. We trained the model for 100 epochs with a learning rate of 0.0005 and used the ADAM optimizer. See Figure 2 for an illustration of the VGG-ish DNN architecture.

3) SIGNAL PROCESSING

Utilizing signal processing, we extracted acoustically meaningful features. In order to choose features appropriate for our task a filtering process is essential having as a criterion their interpretability. If x is the input audio file, we define our

¹<https://github.com/Music-and-Culture-Technology-Lab/omnizart>

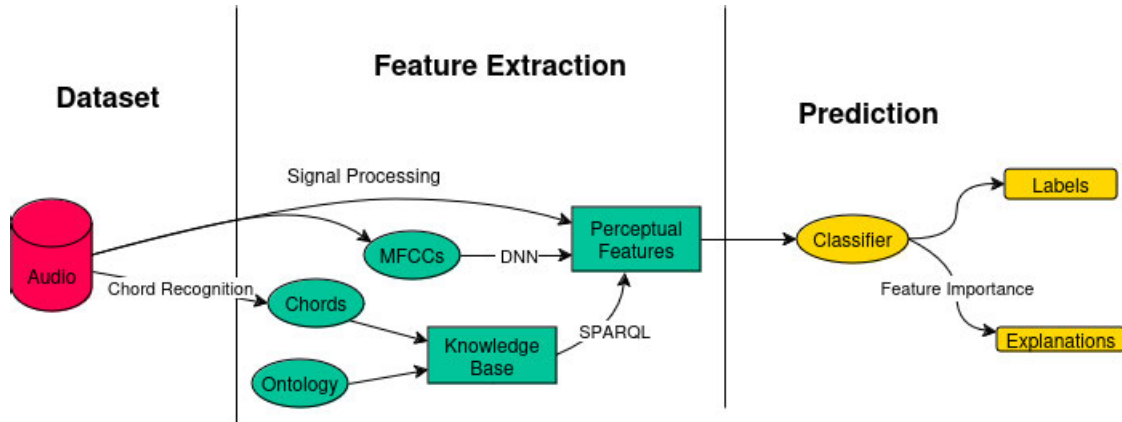


FIGURE 1. Overview of the proposed pipeline.

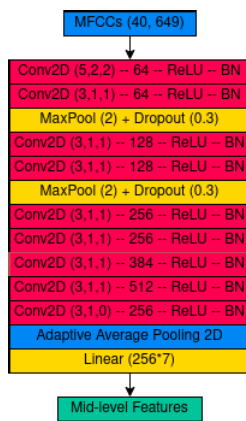


FIGURE 2. The architecture for predicting *midlevel* features.

process as a function E , where $E(x) = x_{\text{signal}}$, $x_{\text{signal}} \in \mathbb{R}^q$ and $q \in \mathbb{N}$ (q the number of the extracted features).

For this approach, we used features extracted with a signal processing tool called Essentia [4]. The features were filtered out from three music experts in order to keep only the ones that are acoustically and musically meaningful. They ended up choosing 23 features:² *Danceability*, *Loudness*, *Chords Changes Rate*, *Dynamic Complexity*, *Zero Crossing Rate*, *Chords Number Rate*, *Pitch Saliency*, *Spectral Centroid*, *Spectral Complexity*, *Spectral Decrease*, *Spectral Energyband High*, *Spectral Energyband-Low*, *Spectral Energyband Middle High*, *Spectral Energyband Middle Low*, *Spectral Entropy*, *Spectral Flux*, *Spectral Rolloff*, *Spectral Spread*, *Onset Rate*, *Length*, *BPM*, *Beats Loud* and *Vocal-Instrumental*. The description of the features could be found in the *GitHub repository*.³ By concatenating x_{harmonic} , x_{midlevel} , x_{signal} , and normalizing the values with a standard scaler, we end up with the feature vectors used to train the models.

²https://essentia.upf.edu/streaming_extractor_music.html#music-descriptors

³<https://github.com/vaslyb/perceptible-music-tagging>

B. INTERPRETABILITY

In our study, we used the extracted features as input to an XGBoost model for music auto-tagging, as described in Section IV-B. We selected XGBoost due to its strong performance and interpretability. To explain our model's predictions, we applied five methods: XGBoost's built-in feature importance, the permutation method, SHAP values, an ablation study, and natural language explanations generated from ChatGPT, prompted with the SHAP values.

1) XGBOOST FEATURE IMPORTANCE

To extract the initial set of feature importance explanations from our XGBoost model, we utilized the model's weights, which indicate the frequency with which each feature appears in the decision trees. By accumulating this information across all tags, we were able to get a general overview of which features were important for the task at hand. Additionally, by viewing the weight only for a specific tag, we were able to gain insight into which features were important for each tag individually.

2) PERMUTATION-BASED FEATURE IMPORTANCE

We computed the second set of feature importance explanations using the permutation-based method. This approach evaluates the contribution of each feature by randomly shuffling its values and measuring the decrease in the model's performance, such as accuracy or loss [7]. A significant drop in performance indicates a high feature importance, while a minor effect suggests a less relevant feature. We used this method to assess which features consistently influenced predictions across all mood tags, helping us better understand the role each feature plays in the model's decision-making process.

3) SHAP

Another method we utilized to extract feature importance was SHAP values [56]. This method estimates how each feature contributes to the prediction based on game theory concepts.

Unlike the weight and permutation methods, SHAP values indicate whether or not features positively or negatively impact predictions. The results are presented as plots sorting features by the sum of SHAP value magnitudes over all samples and showing the distribution of the impacts each feature has on the model output.

4) ABLATION STUDY

For the fourth approach we conducted an ablation study with different groups of features as recommended in the literature [48]. We categorized our features into three groups based on their origin - *Mid-level*, *Harmonic* and *Signal* - and conducted experiments with all possible combinations of the groups, using XGBoost as our classifier. Through these studies, we were able to gain insight into what kind of impact each set of features had on the performance of our best classifier.

5) NATURAL LANGUAGE EXPLANATIONS

To further study the form of the explanations and how they affect the user's understanding, we utilized ChatGPT⁴ to convert explanations generated by SHAP into natural language. It is important to mention that since ChatGPT is itself a opaque it is contradictory to utilize it for generating explanations, and can in some cases even result in misleading the end user (e.g., hallucinations, omitting information etc.). However, we were interested to study how users react to the natural language form of explanations when compared to the other forms.

In particular, we presented figures depicting SHAP values for each tag and prompted ChatGPT with a specific request to explain the model's categorization of a music track under a given tag in a clear, human-understandable manner. The prompt focused on connecting musical concepts to the model's features without explicitly mentioning SHAP and instructed it to incorporate color coding—using pink for high values and blue for low values. This strategy transformed the SHAP explanations into a more coherent and meaningful narrative by exploiting prior knowledge that ChatGPT may have in the music domain [19]. The resulting natural language explanations were also manually checked to ensure that there are no hallucinations or information that is wrong. An example of this procedure is depicted in Figure 3. Notably, a similar methodology for analyzing scientific plots in bioinformatics was applied in a study by [65], which faced analogous challenges. In that work, ChatGPT struggled to distinguish the colors of the plots, a limitation we addressed through further prompting and by providing additional analytical details for the SHAP figure.

IV. EXPERIMENTS

A. DATASETS

The main dataset used for our experiments was the *MTG-Jamendo* dataset [5]. It is an open dataset for music

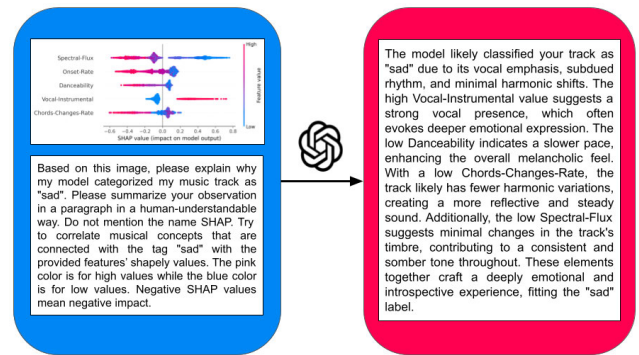


FIGURE 3. An example of prompting ChatGPT to convert SHAP explanations to natural language explanations.

auto-tagging. It was built using music available on the Jamendo platform under Creative Commons licenses and tags provided by content uploaders. The large-scale dataset contains over 18,486 full audio tracks with 56 tags covering mood/theme categories. The extraction of the audio was done by sampling at 22kHz MFCC (40 coefficients) with a window and FFT size of 2048. We opted to utilize this particular dataset due to our intuition that interpretability would prove more valuable in a task involving partially subjective topics, such as mood detection [37].

We utilized the *GTZAN* dataset [64] as our secondary dataset. This collection comprises 1000 audio tracks, each 30 seconds long, featuring 10 distinct genres with 100 tracks per genre. The tracks are uniformly encoded as 16-bit monophonic recordings in .au format, sampled at 22kHz. We selected this dataset due to its widespread use in music genre classification tasks, which provides a well-established benchmark. Additionally, it allowed us to assess the generalizability of our proposed pipeline on a smaller and more controlled corpus, complementing our primary dataset.

For additional evaluation, we employed the *MagnaTagATune* dataset [68]. This dataset consists of 25,863 audio clips, each approximately 30 seconds long, extracted from a total of 5,405 songs released under the Magnatune label.⁵ Alongside the audio files, it includes precomputed acoustic features and tag annotations, which were collected through a gamified approach using the TagATune platform. This platform engaged users in a collaborative task where they labeled music clips based on their listening experience, thereby generating a rich set of annotations for various musical characteristics. The dataset spans a diverse range of genres, including ambient, electronic, classical, and rock. We selected MagnaTagATune due to its extensive use in music tagging research and its unique annotation methodology, which provides valuable insights into human-perceived musical attributes. This allowed us to further validate our approach on a dataset with diverse and user-driven annotations.

⁴OpenAI, ChatGPT, 2024, <https://openai.com/chatgpt>

⁵<http://magnatune.com/>

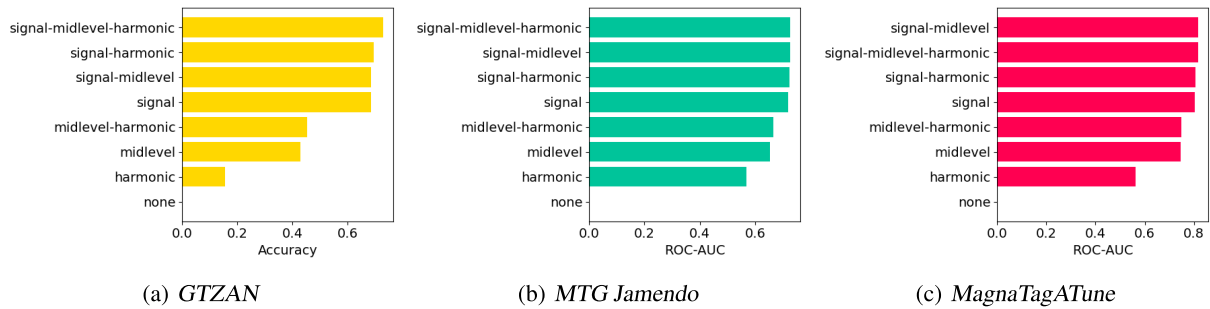


FIGURE 4. Ablation studies. ROC-AUC of different combinations of the six groups of features.

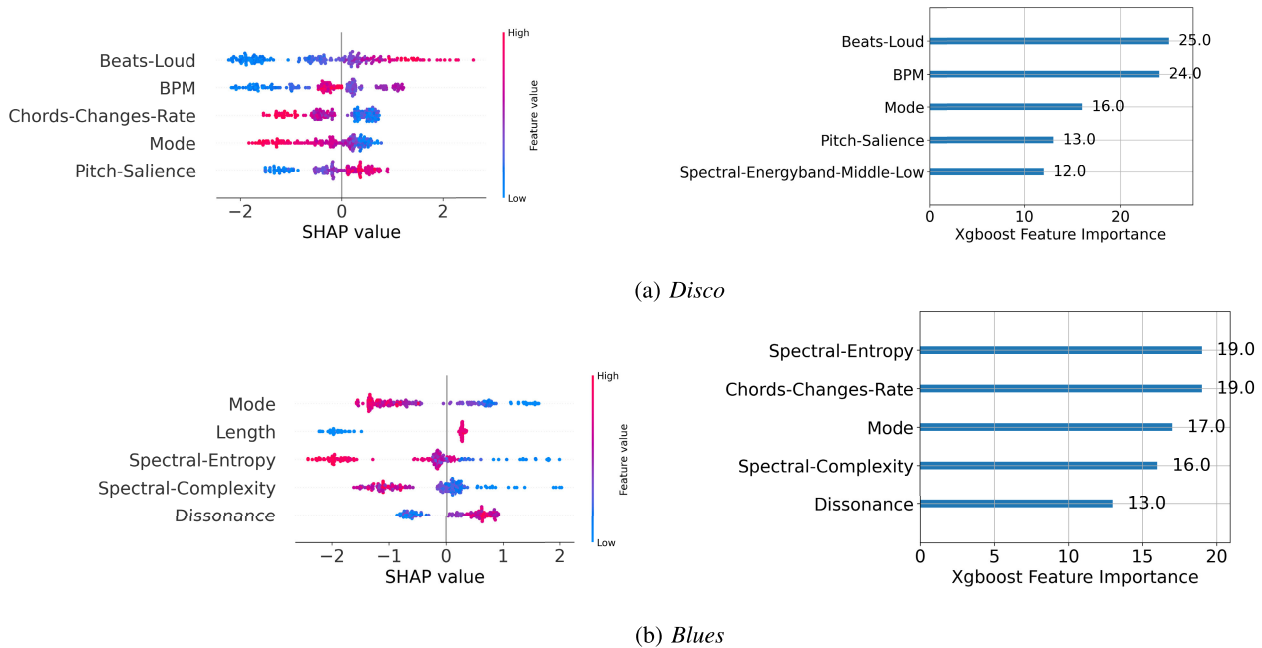


FIGURE 5. SHAP values and XGBoost's feature importance for the predictions of labels *Disco* and *Blues* of the *GTZAN* dataset.

B. RESULTS AND DISCUSSION

1) CLASSIFICATION

After evaluating various traditional machine learning classifiers, we identified XGBoost as the most effective option. XGBoost is an ensemble classifier that leverages decision trees within a gradient-boosting framework. To assess our models, we used the Receiver Operating Characteristic - Area Under the Curve (ROC-AUC) metric, the standard benchmark for performance on the *MTG Jamendo* dataset. Our model achieved an ROC-AUC score of 0.729 on the multi-label mood recognition task with *MTG Jamendo*, surpassing the baseline of 0.725 [62] but falling short of the top-performing intelligent systems, which reached 0.769 [6]. Similarly, in the multi-label classification task with the *MagnaTagATune* dataset, our XGBoost model yielded an ROC-AUC score of 0.84. While slightly below the state-of-the-art performance of 0.91 [16], this result highlights the model's strong predictive capabilities. An overall

TABLE 1. Comparison of our models with state-of-the-art (SOTA) performance.

Dataset	Metric	XGBoost	SOTA
MTG Jamendo	ROC-AUC	0.72	0.76 [6]
MagnaTagATune	ROC-AUC	0.84	0.91 [16]
GTZAN	Accuracy	0.79	0.84 [50]

comparison of our models with state-of-the-art is given in Table 1.

We also obtained promising results when applying XGBoost to the multi-class classification task with the *GTZAN* dataset, achieving an accuracy of 0.79. This performance is slightly below the state-of-the-art accuracy of 0.84 [10], [50]. While our results may not surpass those of advanced deep learning models, the interpretability of our model outputs is valuable for identifying potential biases.

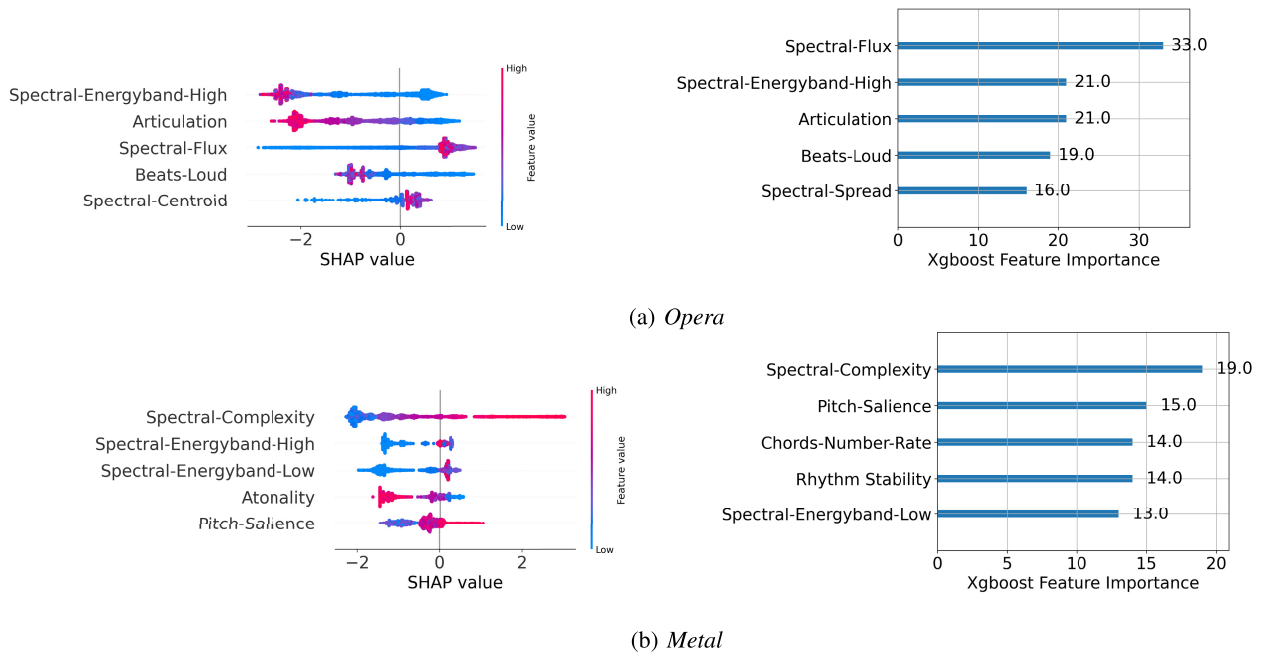


FIGURE 6. SHAP values and XGBoost's feature importance for the predictions of labels *Opera* and *Metal* of the *MagnaTagATune* dataset.

For the collections of perceptual features and the code used for the classification and the interpretation task, we provide the GitHub repository.⁶

2) EXPLAINABILITY

The results of permutation-based and the overall weight XGBoost feature importance for the *MTG-Jamendo* dataset are shown in Fig. 7. The two methods produce similar results, which indicates that the displayed features are likely indeed important. Signal processing features were the most important for predictions, notably *Chords-Changes-Rate*, *Onset-Rate*, *Spectral-Complexity* and *Spectral-Entropy*. The feature *Chords-Changes-Rate* shows how many chord changes exist in a given track, *Onset-Rate* is determined by how many distinguished music notes are heard per minute, while *Spectral-Entropy* and *Spectral-Complexity* are based on the number of peaks in the input spectrum.

We obtained similar results in our ablation study (Fig. 4), where we found that signal processing features were the most useful, while harmonic features were the least useful for both datasets. Considering that harmonic features are extracted from the chord progression, this observation can be attributed to the quality of the chord recognition tool. Additionally, for the *GTZAN* and *MagnaTagATune* datasets, only 30 sec for each track were given, leading to a limited number of chords per track and resulting in not having enough structural information. Consequently, harmonic features had a substantial impact only on predicting the *Blues* labels (see Figure 5), a genre where chord progressions are distinct

and easy to detect throughout a track. In terms of mid-level features, they appeared to be significant in improving the ROC-AUC metric for the *MTG-Jamendo* dataset.

We show examples of interpreting feature importance computed with SHAP values and XGBoost for predicting mood labels in the *MTG Jamendo* dataset. Specifically, we analyze one of the best-performing labels (ROC-AUC > 0.8) and one of the worst-performing mood labels (ROC-AUC = 0.536), as shown in Fig. 11. For the highly-predictable *Christmas* label, we observed that the *Spectral-Energyband-Middle-High* feature has a dominant effect, which makes sense because Christmas songs typically include many instruments with middle and high frequencies, such as triangles and bells. High values on *Dominants* and *Melody* also indicate moving and easy-to-sing songs. Conversely, for the poorly-predictable *Hopeful* mood label (see Figure 11), the fuzzy interpretation of the term and the low performance (ROC-AUC = 0.536) highlight the importance of having well-defined mood labels. We found that both low and high values of *Onset-Rate* affect the negative prediction of the label. Due to the fact that it is considered important by both feature importance methods, negative and positive values should have made a clearer contribution to the final prediction.

For the *MTG Jamendo* dataset, we provide two additional examples illustrating the model's predictions and SHAP feature importance (see Figure 11). The mood predictions for *Trailer* heavily depend on low values of *Length* and high values of *Spectral-Flux*, the latter being a feature used for voice detection. This aligns with the nature of trailers, which are typically shorter than songs and often include

⁶<https://github.com/vaslyb/perceptible-music-tagging>

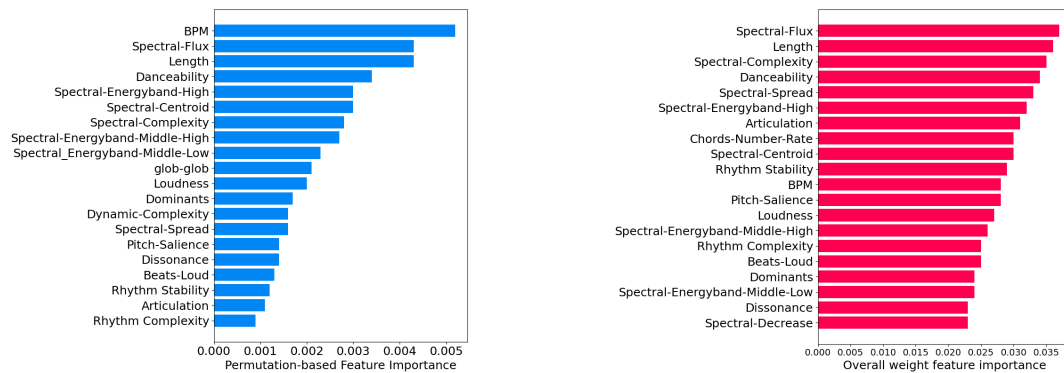


FIGURE 7. Feature importance calculated with two different methods for the MTG Jamendo dataset.

voice-overs. In contrast, for the *Summer* tag, *BPM* emerges as the most significant feature, complemented by high *Onset-Rate* and high values of *Danceability*, all of which are closely associated with dance music. Furthermore, the *Meditative* mood is characterized by low values of *Danceability* and complexity in both rhythm and frequency, along with high values of *Length*, suggesting a preference for calm, simple, and extended compositions.

For *GTZAN*, we show examples of the interpretation for one predicted genre, *Disco* that reached 0.72 F1-score (see Fig. 5). For this label, it is noticeable that high values of *Beat-Loud* and middle values of *BPM* are the most prevalent features. These observations align with the common perception of *Disco* music as a genre characterized by distinguishable beats that are easy to dance to. Overall, SHAP values had a more pronounced impact on the *GTZAN* and *MagnaTagATune* datasets compared to *MTG Jamendo*. We hypothesize that this is because *GTZAN* and *MagnaTagATune* feature more well-defined tags.

For *MagnaTagATune*, we provide two tags with distinctly different timbres: *Opera* and *Metal* (see Fig. 6). We observe that both SHAP values and XGBoost feature importance highlight similar features as the most influential for our model's predictions. Specifically, for *Opera*, we note low values of *Spectral-Energyband-High*, *Articulation*, and *Beats-Loud*, indicating a smooth tone with minimal emphasis on rhythm—characteristic of lyrical opera parts. The presence of *Spectral-Flux* may be attributed to the prominence of human voice in operas. In contrast, for *Metal*, high values of *Spectral-Complexity* suggest a highly dynamic sound. Additionally, the combination of low *Atonality* and high *Pitch-Salience* indicates the presence of a limited set of pitches, which is typical in metal music, where notes and chords often remain unchanged.

As far as the feature *Length* is concerned, it was proven to be a vital feature for the model trained on the *MTG Jamendo* dataset. However, it is critical for our system to not consider this feature only for its predictions. For example, an auto-tagging system might become biased towards assigning the tag *Trailer* to short-length songs (see Fig. 11) and end up

mislabelling samples due to this bias (e.g. when given a short track that is not a *Trailer*). This highlights the importance of interpretability, since knowing that our system considers *Length* for making decisions, allows us to more easily detect and fix such biases which would be impossible in a deep-learning-based setup.

The above examples are meant to illustrate the value of interpretability for the auto-tagging task. It is clear that we can gain insights on a case-by-case basis which would not be possible in an end-to-end setup. Whether or not these insights are worth the additional work needed for feature extraction, and the slight deterioration in performance, depends on each application's priorities. Our argument is that interpretability should be a priority for music auto-tagging systems, as they suffer from label ambiguity, in addition to being at risk of the typical biases of MIR datasets (e.g. over-representation of Western music [53]). To further assess the usability of explainability, we conducted a human survey.

V. HUMAN EVALUATION

A. DESIGNING THE QUESTIONNAIRE

A human evaluation was conducted through a survey to evaluate the methodology and assess the extent to which participants trust the proposed approach to music auto-tagging. This trust was compared to that placed in a state-of-the-art model [6]. The interpretable pipeline is referred to as the Mood XAI model, while the state-of-the-art model is referred to as the Mood AI.

The survey was divided into two parts. In the first part, participants were asked to provide five songs they liked, ensuring diversity in emotion. They were also instructed to indicate 15 seconds of each track. In the second part, we presented tags for these music segments, from which participants could choose as many as they wanted out of a total of nine tags. These nine tags included three randomly selected from the 56 mood tags in the MTG-Jamendo dataset, three representing the top predictions from the Mood XAI, and three derived from the Mood AI.

After participants selected their tags, a music tag suggested by the Mood XAI was presented along with three different

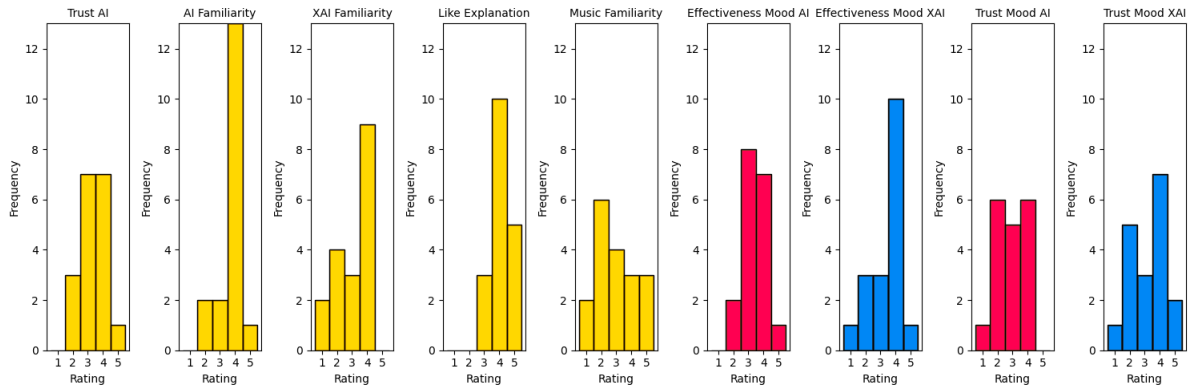


FIGURE 8. The participant's responses to survey questions.

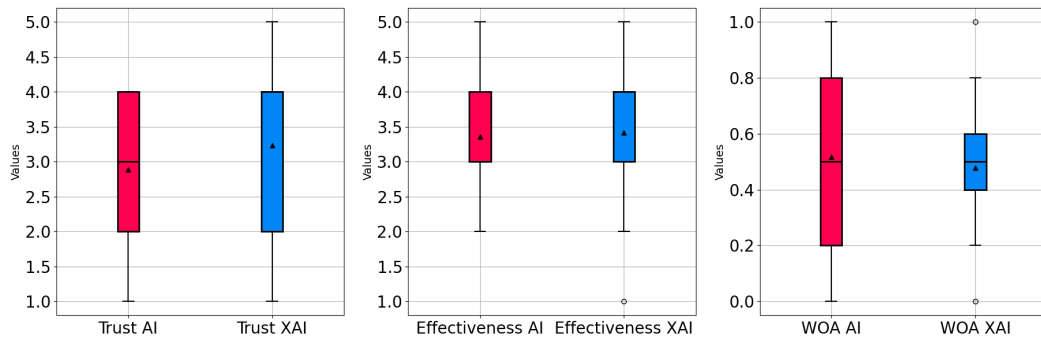


FIGURE 9. Participants trust, effectiveness and WOA in Mood AI and Mood XAI.

types of explanations: SHAP values, feature importance, and natural language explanations. Participants were then asked whether they would incorporate the model's suggestion. This process was repeated for the Mood AI.

Finally, information was gathered on participants' familiarity with music, AI, and XAI. They were asked about their trust in AI, the Mood AI, and the Mood XAI models, as well as their perceptions of the efficiency of both models.

Our approach is based on a Judge-Advisor System (JAS) [57], [58], which features two distinct roles in the decision-making process: the judge and the advisor. The advisor offers suggestions and guidance, while the judge retains sole responsibility for the final decision. In our specific application, Mood AI and Mood XAI serve as the advisor, with the participant assuming the role of the judge, in categorizing the music tracks based on mood.

We measured the trust of the participants in the intelligence systems in two ways. The one involves the *explicit* trust and the other the *implicit* trust. The *explicit* was done with a direct question: *Do you trust Mood XAI?* and a following refinement of the question *If you found it effective, would you still use it to organize your music library?*. The quantification of the *implicit* trust was done with the Weight of Advice (WOA) [31]. WOA is a metric to measure the trust of the participant towards the models and it's calculated with the

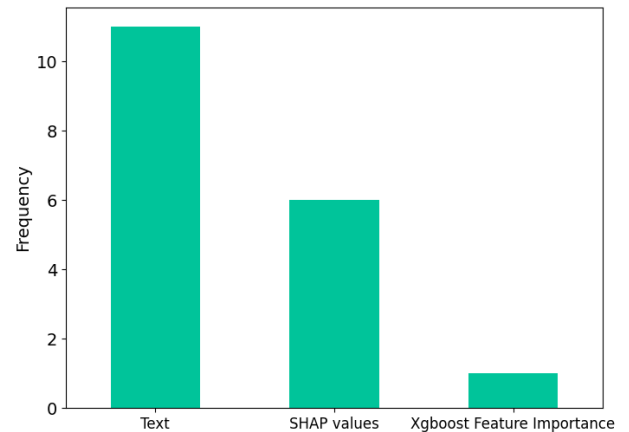
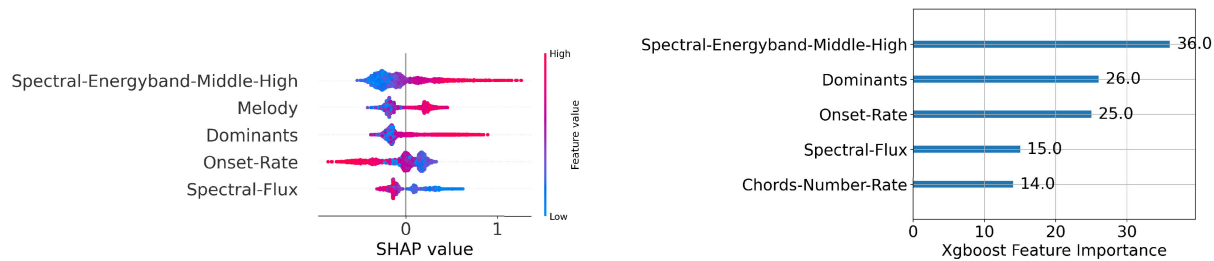


FIGURE 10. The participant's explanation preference.

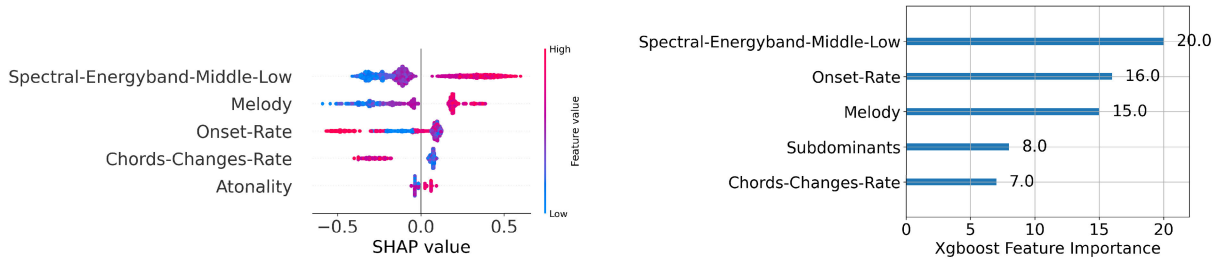
following equation:

$$WOA = \frac{|F - I|}{|A - I|},$$

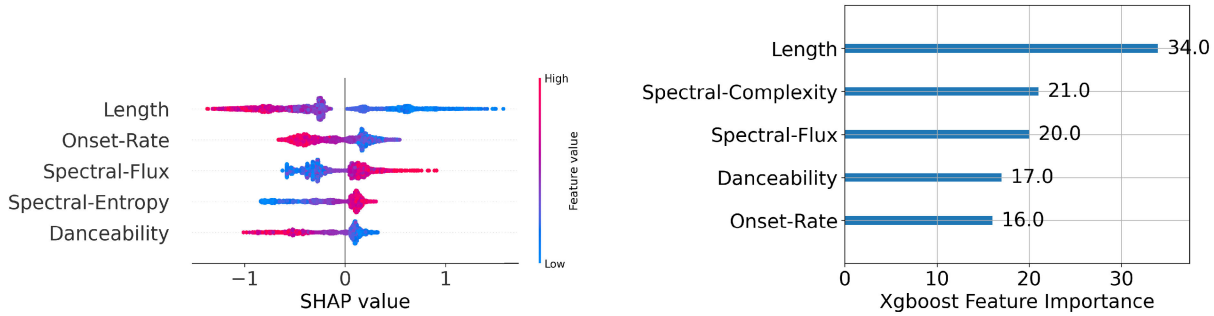
where F and I denote, respectively, the final and initial participant's estimates, while A denotes the algorithmic suggestion.



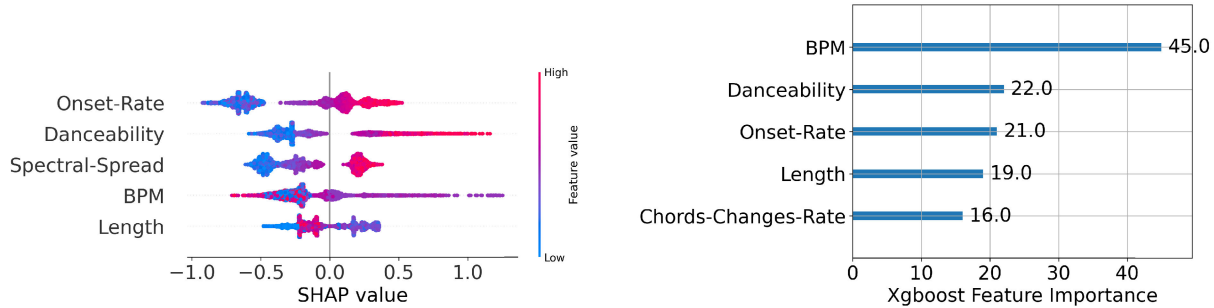
(a) *Christmas*



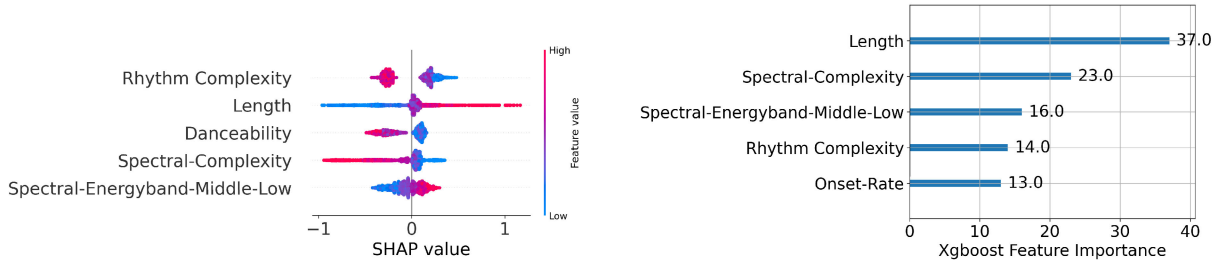
(b) *Hopeful*



(c) *Trailer*



(d) *Summer*



(e) *Meditative*

FIGURE 11. SHAP values and XGBoost's feature importance for the predictions of labels *Christmas*, *Hopeful*, *Trailer* and *Summer* of MTG Jamendo dataset labels.

To assess the preference of the participants in the models (*Effectiveness*), we employed two approaches: an *explicit* method and an *implicit* method. The *explicit* method involved asking participants a direct question: *Was Mood AI effective in categorizing music tracks based on mood?*. In contrast, the *implicit* method assessed accuracy by calculating the proportion of proposed tags from our models that participants initially selected.

You can find examples of both parts of the questionnaire linked below. The first part is available here,⁷ while the second part can be accessed here.⁸

B. RESULTS

A total of 18 participants, consisting of 12 males and 6 females, completed the survey, being informed that their answers would be part of the research. As illustrated in Figure 8, the responses indicate that participants generally trust AI without necessarily having prior experience with it. Interestingly, although many participants were not well-acquainted with XAI, they expressed a positive reception to the explanations offered.

Participants' trust and preference for Mood AI and Mood XAI reveal nuanced differences. As shown in Figure 9, there is a slight preference for Mood XAI in terms of explicit trust (*Trust*), a small preference for Mood AI as far as the performance is concerned (*Effectiveness*), and an equal preference for both models when it comes to willingness to accept advice (*WOA*). Implicit efficiency results, measured by the proportion of tags suggested by our models that participants initially chose, indicate that Mood AI achieved a higher accuracy (0.4) compared to Mood XAI (0.15), suggesting that participants implicitly relied more on Mood AI. Overall, there are only minor differences in trust and preference between Mood AI and Mood XAI, suggesting that despite the higher accuracy of the state-of-the-art model, users tend to rely on other aspects when evaluating an intelligent system. These results align with previous studies, which found that explanations increase user trust [21], [33].

To assess the agreement of our observed variables (*Trust*, *WOA*, *Effectiveness*), we computed Cronbach's Alpha, a reliability coefficient used to evaluate the internal consistency of a set of items. The analysis yielded values of 0.720 for Mood AI and 0.764 for Mood XAI, both of which suggest acceptable levels of reliability. These results indicate that the selected variables are sufficiently correlated, supporting the hypothesis that they measure the same latent variable—user trust in the system. This consistency is crucial for ensuring the robustness of our findings across different conditions.

In terms of participants' preferences for the type of explanations, Figure 10 shows a clear preference for Natural Language Explanations. This outcome is on the one hand expected, as many participants lacked a strong background in music, making it difficult for them to grasp

feature-importance-based explanations, but on the other hand might be alarming, since ChatGPT lacks the rigour of the other tested forms of explanations, thus potentially resulting in misleading users, which is especially problematic if this is the format that users so clearly prefer. The users' preference was further reflected in the freeform comments we collected from participants. Several participants mentioned that they did not understand the differences between spectral values, while others noted that the explanations were too technical and suggested they should be more intuitive. This highlights an emerging challenge in explainable AI in general, where LLM-based explanations are more familiar and intuitive to users, thus increasing trust, while in tandem they are less transparent and have no guarantee of accuracy, thus increasing the potential for misinforming users.

VI. CONCLUSION AND FUTURE WORK

We present a novel music auto-tagging pipeline designed to infer mood tags from audio with a focus on interpretability. In addition, we introduce a new set of harmony-based music features that enhance the prediction process. Our research underscores the importance of feature engineering in developing human-understandable, trustworthy representations that not only boost model performance but also improve transparency. Furthermore, we discuss potential biases in the model and data, exploring how different types of explanations shed light on these issues.

The pipeline underwent evaluation through a user survey, where participants were asked to assess their trust in the system. While the state-of-the-art model outperformed the interpretable pipeline across several performance metrics, users reported comparable levels of trust in both systems. This suggests that while advanced models may achieve higher technical performance, the interpretability of features can bridge the gap in terms of user confidence and trust.

In future work, we aim to explore a range of explanation methods and assess their effectiveness across diverse user groups. The feature extraction process will also be expanded to include a richer set of music theory-based features, along with the integration of additional modalities, such as lyrics. Another important direction will be to investigate inherently interpretable algorithms, such as *Group Lasso* or the *EMBanded* algorithm, to further enhance music auto-tagging transparency.

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⁷<https://forms.gle/G4T5VYXvMJvuMxiZ6>

⁸<https://forms.gle/nXyG3XVxn5ia689L7>

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